

Lynbrook Math Summer Camp 2026

N4 Base Systems

Lynbrook Math Club

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N4.1 Lecture

Most of the numbers we write use the decimal, or base-10, system. For example,

$$4726 = 4 \cdot 10^3 + 7 \cdot 10^2 + 2 \cdot 10 + 6.$$

However, there is nothing mathematically special about using powers of 10. Computers commonly use base 2, and many contest problems ask us to interpret numbers written in unfamiliar bases.

For example, what does

$$(2314)_5$$

mean? How can we quickly convert

$$(110101)_2$$

to decimal? Can a numeral have the same value in two different bases? Base systems allow us to answer these questions systematically.

Place Value in an Arbitrary Base

Definition N4.1.1 (Base- b Representation). Let $b \geq 2$ be an integer. A base- b numeral

$$(a_n a_{n-1} \cdots a_1 a_0)_b$$

represents the integer

$$a_n b^n + a_{n-1} b^{n-1} + \cdots + a_1 b + a_0,$$

where each digit satisfies

$$0 \leq a_i \leq b - 1.$$

Thus, the allowed digits in base b are

$$0, 1, 2, \dots, b - 1.$$

For example, in base 5, the allowed digits are 0, 1, 2, 3, 4. Therefore, $(2431)_5$ is valid, but $(2531)_5$ is not.

Theorem N4.1.2 (Place-Value Formula)

Let $b \geq 2$, and let $a_0, a_1, \dots, a_n$ be digits satisfying

$$0 \leq a_i < b.$$

Then

$$(a_n a_{n-1} \cdots a_1 a_0)_b = \sum_{i=0}^n a_i b^i.$$

This theorem is simply the formal statement of what a base- b numeral means.

Example 3

Convert $(2314)_5$ to base 10.

Using place value,

$$(2314)_5 = 2 \cdot 5^3 + 3 \cdot 5^2 + 1 \cdot 5 + 4.$$

Therefore,

$$(2314)_5 = 250 + 75 + 5 + 4 = \boxed{334}.$$

Example 4

Convert $(110101)_2$ to base 10.

$$\begin{aligned} (110101)_2 &= 1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^2 + 1 \\ &= 32 + 16 + 4 + 1 \\ &= \boxed{53}. \end{aligned}$$

Horner's Method

Instead of calculating every power separately, we may evaluate a numeral from left to right.

Theorem N4.1.5 (Horner's Method)

For a base- b numeral

$$(a_n a_{n-1} \cdots a_1 a_0)_b,$$

we have

$$(a_n a_{n-1} \cdots a_1 a_0)_b = (\cdots ((a_n b + a_{n-1})b + a_{n-2})b + \cdots)b + a_0.$$

Proof. Starting with the right-hand side and distributing repeatedly gives

$$a_n b^n + a_{n-1} b^{n-1} + \cdots + a_1 b + a_0,$$

which is exactly the place-value formula. □

For example,

$$(2314)_5 = ((2 \cdot 5 + 3) \cdot 5 + 1) \cdot 5 + 4.$$

Example 6

Evaluate $(43201)_7$ using Horner's method.

Compute successively:

$$\begin{aligned} 4 \cdot 7 + 3 &= 31, \\ 31 \cdot 7 + 2 &= 219, \\ 219 \cdot 7 + 0 &= 1533, \\ 1533 \cdot 7 + 1 &= 10732. \end{aligned}$$

Thus,

$$\boxed{(43201)_7 = 10732}.$$

Converting from Base 10

Theorem N4.1.7 (Repeated-Division Conversion)

Let N be a positive integer and let $b \geq 2$. Repeatedly divide by b :

$$\begin{aligned} N &= bq_0 + r_0, \\ q_0 &= bq_1 + r_1, \\ q_1 &= bq_2 + r_2, \\ &\vdots \end{aligned}$$

where

$$0 \leq r_i < b.$$

When the quotient eventually becomes 0, the base- b representation of N is obtained by reading the remainders from bottom to top.

Proof. The first equation gives

$$N = bq_0 + r_0,$$

so r_0 is the units digit.

Substituting

$$q_0 = bq_1 + r_1$$

gives

$$N = b(bq_1 + r_1) + r_0 = b^2q_1 + r_1b + r_0.$$

Continuing this substitution produces

$$N = r_kb^k + \cdots + r_1b + r_0,$$

so the digits are $r_k, \dots, r_1, r_0$. □

Example 8

Convert 156 to base 5.

$$156 = 5(31) + 1,$$

$$31 = 5(6) + 1,$$

$$6 = 5(1) + 1,$$

$$1 = 5(0) + 1.$$

Reading the remainders from bottom to top gives

$$\boxed{156 = (1111)_5}.$$

Remark N4.1.9. The first remainder is the units digit because

$$N = bq + r, \quad 0 \leq r < b.$$

Repeating the process on q finds the remaining digits.

Comparing Numbers in the Same Base

Theorem N4.1.10 (Comparison of Base- b Numerals)

Suppose two positive integers are written in the same base b .

1. The numeral with more digits represents the larger integer.
2. If the numerals have the same number of digits, compare their digits from left to right. At the first position where they differ, the numeral with the larger digit is larger.

Proof. A base- b numeral with $n + 1$ digits is at least

$$b^n,$$

while every numeral with at most n digits is less than

$$b^n.$$

This proves the first statement.

For the second statement, the first differing digit multiplies a larger power of b than all later digits. Its contribution cannot be overcome by the remaining lower-place digits. $\square$

For example,

$$(10000)_6 > (5555)_6$$

because the first numeral has more digits, and

$$(3412)_7 > (3366)_7$$

because the first differing digits are 4 and 3.

Addition in Other Bases

Addition works as in base 10, except that we carry whenever a column total reaches the base.

Theorem N4.1.11 (Carrying in Base b)

Suppose a column in a base- b addition has total T . Write

$$T = bq + r, \quad 0 \leq r < b.$$

Then r is written in the current column, and q is carried to the next column.

This is a direct application of the division algorithm.

Example 12

Compute

$$(243)_5 + (134)_5.$$

In the units column,

$$3 + 4 = 7 = 5(1) + 2,$$

so write 2 and carry 1.

In the next column,

$$4 + 3 + 1 = 8 = 5(1) + 3,$$

so write 3 and carry 1.

Finally,

$$2 + 1 + 1 = 4.$$

Therefore,

$$(243)_5 + (134)_5 = (432)_5.$$

Example 13

Compute

$$(11011)_2 + (1011)_2.$$

Aligning the digits gives

$$\begin{array}{r} 11011 \\ 01011 \\ \hline 100110 \end{array}$$

so

$$(11011)_2 + (1011)_2 = (100110)_2.$$

Subtraction and Multiplication

Subtraction works as usual, except that borrowing 1 from the next column adds b to the current column. Multiplication also works as usual, with all carrying performed in the chosen base.

Example 14

Compute

$$(4032)_5 - (1244)_5.$$

Borrowing and subtracting in base 5 gives

$$(4032)_5 - (1244)_5 = (2233)_5.$$

Example 15

Compute

$$(23)_5(14)_5.$$

Direct multiplication in base 5 gives

$$\begin{array}{r} 23 \\ \times 14 \\ \hline 202 \\ 230 \\ \hline 432 \end{array}$$

Therefore,

$$(23)_5(14)_5 = (432)_5.$$

As a check,

$$(23)_5 = 13, \quad (14)_5 = 9,$$

and

$$13 \cdot 9 = 117 = (432)_5.$$

Binary, Octal, and Hexadecimal

Base 2 is called **binary**; its only digits are 0 and 1. Base 8 is called **octal**. Base 16 is called **hexadecimal**, where

$$A, B, C, D, E, F$$

represent

$$10, 11, 12, 13, 14, 15.$$

These bases are compatible because

$$8 = 2^3 \quad \text{and} \quad 16 = 2^4.$$

Theorem N4.1.16 (Grouping Binary Digits)

Let k be a positive integer. To convert a binary numeral to base 2^k , group its binary digits into blocks of k , beginning from the right. Each block becomes one base- 2^k digit.

Proof. A block of k binary digits represents an integer from

$$0 \quad \text{through} \quad 2^k - 1,$$

so it is exactly one valid base- 2^k digit.

Also, shifting one block to the left multiplies its value by 2^k , which is the place-value multiplier in base 2^k . $\square$

Example 17

Convert $(110101011)_2$ to base 8.

Since

$$8 = 2^3,$$

group the binary digits into blocks of 3:

$$110 \quad 101 \quad 011.$$

These blocks represent 6, 5, 3, so

$$(110101011)_2 = (653)_8.$$

Example 18

Convert $(101111001101)_2$ to base 16.

Since

$$16 = 2^4,$$

group into blocks of 4:

$$1011 \quad 1100 \quad 1101.$$

These blocks represent B, C, D , so

$$(101111001101)_2 = (BCD)_{16}.$$

Unknown Bases

Contest problems often give a numeral while leaving the base unknown.

Theorem N4.1.19 (Unknown-Base Method)

To solve an equation involving an unknown base b :

1. Expand each numeral using place value.
2. Solve the resulting algebraic equation.
3. Keep only integer solutions satisfying

$$b \geq 2$$

and

$b > \text{every digit appearing in the numeral.}$

Example 20

In what base does

$$(132)_b = 44$$

hold?

By place value,

$$(132)_b = b^2 + 3b + 2.$$

Thus,

$$b^2 + 3b + 2 = 44,$$

so

$$b^2 + 3b - 42 = 0.$$

Factoring,

$$(b + 7)(b - 6) = 0.$$

The possible algebraic solutions are

$$b = -7 \quad \text{and} \quad b = 6.$$

A base must be positive and larger than every digit used, so

$$\boxed{b = 6}.$$

Remark N4.1.21. Always check that the base is larger than every digit appearing in the numeral. A numeral containing the digit 7 must be written in base at least 8.

Repeated Digits

Theorem N4.1.22 (Repeated Ones)

For integers $b \geq 2$ and $n \geq 1$,

$$\underbrace{(11 \cdots 1)}_{n \text{ digits}}_b = 1 + b + b^2 + \cdots + b^{n-1} = \frac{b^n - 1}{b - 1}.$$

Proof. By place value,

$$(\underbrace{11 \cdots 1}_n)_b = 1 + b + b^2 + \cdots + b^{n-1}.$$

This is a geometric series with first term 1 and common ratio b , so

$$1 + b + \cdots + b^{n-1} = \frac{b^n - 1}{b - 1}.$$

□

Theorem N4.1.23 (Repeated Digit Formula)

Let d be a valid base- b digit. Then

$$(\underbrace{dd \cdots d}_n)_b = d \frac{b^n - 1}{b - 1}.$$

Proof. By place value,

$$(\underbrace{dd \cdots d}_n)_b = d(1 + b + \cdots + b^{n-1}).$$

Apply the Repeated Ones Theorem. □

Example 24

Evaluate $(11111)_3$.

$$(11111)_3 = \frac{3^5 - 1}{3 - 1} = \frac{243 - 1}{2} = \boxed{121}.$$

Trailing Zeros

Theorem N4.1.25 (Trailing Zeros in Base b)

Let N be a positive integer. Its base- b representation ends in at least k zeros if and only if

$$b^k \mid N.$$

It ends in exactly k zeros if and only if

$$b^k \mid N \quad \text{and} \quad b^{k+1} \nmid N.$$

Proof. Ending in at least k zeros means that

$$N = (a_n a_{n-1} \cdots a_k \underbrace{00 \cdots 0}_k)_b.$$

Every nonzero place value is then a multiple of b^k , so

$$b^k \mid N.$$

Conversely, if $b^k \mid N$, then the remainder of N upon division by b^k is 0. Therefore, its final k base- b digits are all 0. □

Example 26

How many trailing zeros does 3600 have in base 6?

Since

$$3600 = 6^2 \cdot 100,$$

we have

$$6^2 \mid 3600.$$

However,

$$6^3 \nmid 3600.$$

Therefore, the base-6 representation of 3600 ends in exactly

$$\boxed{2}$$

zeros.

Last Digits and Modular Arithmetic

Theorem N4.1.27 (Last- k -Digits Principle)

Let

$$N = (a_n a_{n-1} \cdots a_1 a_0)_b.$$

Then, for every positive integer k ,

$$N \equiv a_{k-1}b^{k-1} + \cdots + a_1b + a_0 \pmod{b^k}.$$

In other words, the last k base- b digits determine the remainder modulo b^k .

Proof. The contribution from every digit to the left of the final k digits contains a factor of b^k . Therefore, those terms are all congruent to 0 modulo b^k . $\square$

In particular,

$$N \equiv a_0 \pmod{b}$$

and

$$N \equiv a_1b + a_0 \pmod{b^2}.$$

Divisibility Tests in Base b

Theorem N4.1.28 (Digit-Sum Test in Base b)

Let

$$N = (a_n a_{n-1} \cdots a_1 a_0)_b.$$

Then

$$N \equiv a_n + a_{n-1} + \cdots + a_1 + a_0 \pmod{b-1}.$$

Therefore, N is divisible by $b-1$ if and only if its digit sum is divisible by $b-1$.

Proof. Since

$$b \equiv 1 \pmod{b-1},$$

we have

$$b^j \equiv 1 \pmod{b-1}$$

for every nonnegative integer j . Hence,

$$\begin{aligned} N &= a_nb^n + a_{n-1}b^{n-1} + \cdots + a_1b + a_0 \\ &\equiv a_n + a_{n-1} + \cdots + a_1 + a_0 \pmod{b-1}. \end{aligned}$$

□

For base 10, this is the usual divisibility test for 9.

Theorem N4.1.29 (Alternating-Digit-Sum Test in Base b)

Let

$$N = (a_na_{n-1}\cdots a_1a_0)_b.$$

Then

$$N \equiv a_0 - a_1 + a_2 - a_3 + \cdots + (-1)^n a_n \pmod{b+1}.$$

Therefore, N is divisible by $b+1$ if and only if its alternating digit sum is divisible by $b+1$.

Proof. Since

$$b \equiv -1 \pmod{b+1},$$

we have

$$b^j \equiv (-1)^j \pmod{b+1}.$$

Substituting into the place-value expansion gives

$$N \equiv a_0 - a_1 + a_2 - a_3 + \cdots + (-1)^n a_n \pmod{b+1}.$$

□

For base 10, this is the usual divisibility test for 11.

Example 30

Determine whether $(243615)_7$ is divisible by 6.

Since

$$6 = 7 - 1,$$

we use the digit-sum test:

$$2 + 4 + 3 + 6 + 1 + 5 = 21.$$

Because

$$6 \nmid 21,$$

the number is not divisible by 6.

Palindromes

Definition N4.1.31 (Palindrome). A numeral is a **palindrome** if it reads the same forward and backward.

Theorem N4.1.32 (Even-Length Palindromes)

Every even-length palindrome in base b is divisible by $b + 1$.

Proof. Consider an even-length palindrome

$$(a_1 a_2 \cdots a_k a_k \cdots a_2 a_1)_b.$$

Its alternating digit sum is 0, because matching digits occur with opposite signs and cancel in pairs.

By the alternating-digit-sum test, the numeral is divisible by

$$b + 1.$$

□

Remark N4.1.33. In base 10, this proves that every even-length decimal palindrome is divisible by 11.

Counting Numerals

Theorem N4.1.34 (Number of n -Digit Base- b Numerals)

For integers $b \geq 2$ and $n \geq 1$, the number of positive n -digit base- b numerals is

$$(b - 1)b^{n-1}.$$

Proof. The first digit has

$$b - 1$$

choices because it may be any digit except 0. Each of the remaining $n - 1$ digits has

$$b$$

choices. By the multiplication principle, the total is

$$(b - 1)b^{n-1}.$$

□

Example 35

How many three-digit base-7 numerals have no repeated digits?

The first digit has 6 choices because it cannot be 0.

The second digit has 6 remaining choices, since it may be 0 but may not repeat the first digit.

The third digit has 5 remaining choices. Therefore,

$$6 \cdot 6 \cdot 5 = \boxed{180}.$$

Example 36

How many positive integers less than $(10000)_3$ have base-3 representations using only the digits 0 and 1?

Every such integer may be represented by a string of four digits, allowing leading zeros. Each digit has two choices:

$$0 \quad \text{or} \quad 1.$$

Thus, there are

$$2^4 = 16$$

possible strings.

The all-zero string represents 0, which is not positive. Therefore, the answer is

$$\boxed{15}.$$

Useful Contest Patterns

- Convert a numeral into an algebraic expression in its base.
- Use repeated division to convert from decimal.
- Check that every digit is smaller than the base.
- Interpret repeated digits using a geometric series.
- Use the final k digits to work modulo b^k .
- Use digit sums modulo $b - 1$.
- Use alternating digit sums modulo $b + 1$.
- Group binary digits when converting to base 8 or base 16.
- Translate unknown-base problems into polynomial equations.
- Count numerals by treating digit positions as choices.